

RESUMPTION

Helix OTA — Session Resumption (canonical, always-current)

Field	Value
Revision	9
Last modified	2026-06-10T16:30:00Z
Status	active — the §11.4.131 single canonical out-of-the-box session-resumption file
Standard path	docs/RESUMPTION.md (this file — the fixed project-declared §11.4.131 entry point; do not move without a §11.4.66 operator decision) Point any fresh session at THIS file. It carries a SHORT one-line resume + a FULL block, the read-first handoff docs, exact live-state anchors, current PHASE + NEXT + terminal goal, and the binding constraints. Moment-valid for HEAD eb9e1c4 on main (all 5 remotes aligned). Overnight autonomous re-validation 2026-06-10 (operator away-from-keyboard): EVERY achievable software tier RE-RUN fresh on eb9e1c4 and GREEN — Go build/vet/race, pgx integration, inheritance gate, meta-test §1.1, dashboard 58/58, all Go submodule cores, HelixQA LIVE 10/0 + self-test, podman full-lifecycle + fleet 5/5 + recall-recovery 7/7, QEMU firmware. See docs/qa/STABILITY_REPORT.md Rev 2. Android bricks product-identical to proof (pins 1061015/8bb8d2f unchanged), not re-run (§12.6/§11.4.6). BLOCKED (hardware): T2 Cuttlefish / GSI-A·B real-apply / T3 RK3588 → no release tag (§11.4.40 would be a bluff).
Status summary	

SHORT — paste this first sentence into a fresh session

Read docs/research/main_specs/CONTINUATION.md first, run `git fetch --all --prune`, then continue the Helix OTA autonomous loop toward the next validated-and-published version tag — current PHASE is “OVERNIGHT STABLE — every achievable tier RE-VERIFIED GREEN on this HEAD (see docs/qa/STABILITY_REPORT.md Rev 2); P4 Cuttlefish + GSI-A/B + T3 RK3588 remain hardware-BLOCKED so no release tag yet (§11.4.40)”; HEAD is eb9e1c4 on main (all 5 remotes aligned).

FULL — detailed resumption block

0. Read FIRST (in order)

1. docs/research/main_specs/CONTINUATION.md — the live work-state handoff (DONE / NEXT / locked decisions). **This is the primary handoff doc.**
2. .remember/remember.md — present but currently empty; no extra state there.
3. docs/design/EMULATED_DEVICE_TESTING.md — the tiered emulator-testing plan (Tier-1/2/3) now in flight.
4. Then run: `git fetch --all --prune` and reconcile HEAD. .@{u} before any edit (§11.4.37 fetch-before-edit).

1. Exact live-state anchors (moment-valid 2026-06-10)

- **HEAD commit:** eb9e1c4 — test(evidence): refresh bank evidence_artifacts from determinism run-2 (on main, all 5 remotes aligned; the overnight re-validation + doc-sync lands on top). **Build state: GREEN — re-verified fresh on THIS HEAD; see docs/qa/STABILITY_REPORT.md Rev 2.** NOTE (§11.4.6 fact-correction): weasyprint 66.0 + pandoc ARE present on this host — the prior “no weasyprint/LaTeX → PDF gap” note is stale; PDF siblings can be generated via `scripts/export_docs.sh`.
- **Wave 6 shipped this session (2026-06-10, on main, all 4 upstreams):**
 - 16b4b23 android gitlink → ota-android-agent 26bd0a2 (:android AAR now builds — root-caused to an unpinned kotlin.android plugin, NOT MPP; :core 47/0 green).
 - 12087bc emulator failure→recall(forward-fix)→recovery e2e (TestRecallRecoveryE2E PASS +-race; podman shell variant tests/emulator/tier1_recall_recovery_e2e.sh syntax-clean, **conductor must run it live**).
 - 19e1346 dashboard §11.4.135 regression guard for the DeviceStatus.health-object crash (Vitest 58, RED/GREEN polarity).
 - 3c85b14 security probes recall+telemetry 28/0 (surfaced a STRONGER trust boundary — strict bindJSON rejects injected keys 400).
 - 88bd2c2 recall lifecycle e2e live 35/0 (tests/e2e/recall_lifecycle.sh).
 - 5123e67 HelixQA bluff audit + real bank runner (tools/helixqa/run_bank.sh): VERDICT HelixQA is NOT a bluff — canonical engine gates on dispatch-exit-0 + non-empty evidence ledger + catches its own

negation; the real gap was an INCORPORATION gap (§11.4.27, never wired as a submodule / bank never machine-executed). Runner: dry-run 10/0/0 + self-test PASS. Audit: docs/research/helixqa_bluff_audit/REPORT.md.

- 71be1cd **per-event telemetry duration_ms+bytes_transferred end-to-end** (closed the parked WIDEN row-4): ota-protocol gitlink → 3d360ab (additive, validation rejects negative), emulator emits → server ingest → store (memory+pgx) → read view; TestEmulatorTelemetryFieldsEndToEnd exact-value PASS + real-Postgres parity (TestPostgresRepositoryContract_Integration, podman PG, 0 skips).
- 8a03a63+de87e21 **emulation test-fabric research + full design** (docs/research/emulation_infra/REPORT.md cited; docs/design/emulation_fabric/{DESIGN,ROADMAP,TEST_COVERAGE_PLAN}.md+SCHEMA.sql) — tiers T0–T3/Tfw/Tcp, extends containers submodule, honest macOS-M3 KVM gaps.
- **Queue done this session (all pushed, all 4 upstreams):** ✓ HelixQA LIVE full-bank 10/0 (d5fcf4a — fixed run_bank LIVE-mode <pw> substitution + free-port isolation); ✓ podman recall→recovery e2e 7/7 (71c3f77); ✓ **HelixQA wired as a submodule** at submodules/helixqa pinned latest fe17c08, gate Part C PASS, install_upstreams done (d488e6e, §11.4.27); ✓ **emulation P1 T1-tier AVD-HVF boot smoke** arm64-v8a booted+accel-proven (5d6b3ae).
- **Wave 7 (5-agent parallel, all pushed):** ✓ **fabric target-registry** project-side Go (c709c92 — server/internal/fabric + store memory+pgx, exclusive-lease §11.4.119 + non-empty-evidence §11.4.69, real-PG integration 0 skips, §1.1 mutation); ✓ **on-device AVD instrumentation test** → ota-android-agent 1061015 (8266355 gitlink; OK 5 tests 3x; UpdateEngine-present-but-degrades-to-Failed FACT); ✓ **P2 Tfw QEMU smoke** authored, honest SKIP (qemu absent — brew install qemu to activate) (504c92b); ✓ **helix-deps.yaml §11.4.31** + native test_cases bank (607e7c3); ✓ spec_impl_alignment row-4 RESOLVED + coverage_ledger Rev2 (7a7cb48).
- **Overnight stability hardening (2026-06-10, all pushed):** ✓ **P2 QEMU firmware tier ACTIVATED** — brew install qemu (11.0.1), tier_fw_qemu_smoke.sh real edk2 UEFI boot PASS (93ac0a3); ✓ **run_bank —self-test wired into the pre-build gate** (60da61a, §1.1 every build); ✓ **Challenges submodule incorporated** at submodules/challenges (438057d, §11.4.27, gate PASS, helix-deps 11 deps); ✓ **comprehensive stability sweep ALL GREEN** on HEAD 438057d (Go all-tiers+race, pgx integration, meta-test, dashboard 58, HelixQA LIVE bank 10/0, podman full-lifecycle + fleet 5/5 + recall-recovery, QEMU tier, AVD boot + on-device, submodule cores) — see docs/qa/STABILITY_REPORT.md. DEFERRED zero-risk: fabric scheduler P3 (would be unwired §11.4.124), HelixQA in-tree compile (./containers layout, HelixQA-side), PDF doc siblings (no weasyprint/LaTeX on host).
- **Remaining NEXT (real PWUs):** (1) **emulation P4 Cuttlefish** (Linux+KVM) for real update_engine A/B + the **GSI-A/B real-apply** question (REPORT §2.2/§8 — on a google_apis AVD UpdateEngine is present but unusable for a non-system caller); (2) **P2 QEMU activate** (brew install qemu → tier_fw_qemu_smoke.sh goes green); (3) **P3 fabric scheduler** wiring on top of the registry (Nomad/LAVA backend); (4) wire HelixQA's missing challenges submodule (its go.mod replace digital.vasic.challenges points at an absent submodules/challenges); (5) HelixQA: run_bank.sh —self-test into the pre-build gate, Dispatcher→autonomous-runner (§11.4.124); (6) PDF

export pipeline on host (weasyprint) so §11.4.65 .pdf siblings stay in sync (currently HTML-only, LaTeX/weasyprint absent).

- **Wave 5 shipped (on main):** 2391cb6 **full OTA lifecycle + multi-device fleet PROVEN on podman** (single device 1.0.0→1.1.0 with the complete download→success telemetry accepted; fleet 5/5 → 1.0.0→1.2.0; evidence docs/qa/20260610T11191*/ + ...111928Z-fleet/); f5a3428 N-device scaling (50 concurrent, ~1580 ops/sec, 0 err, -race); 4658125 dashboard Fleet-detail + a real product bug fix (DeviceStatus.health object). **The emulator now fully runs+tests the codebase end-to-end on podman — the prior “honest gap” is closed.**
- **NEXT non-blocked candidates:** richer telemetry ingest fields (duration_ms/ bytes_transferred — needs the device to send them); a rollback/recall lifecycle e2e on the emulator (drive a failure event → recall forward-fix); migrate the dashboard health-bug to a Fixed.md ATM entry. **BLOCKED:** ATM-003 Tier-2 (Linux+KVM Cuttlefish, design ready in docs/design/CUTTLEFISH_TIER2.md) + ATM-004 Tier-3 (RK3588 hardware).
- **Waves 3–4 also shipped (on main):** 3c57867 closed both protocol gaps (UpdateAvailable.deployment_id [ota-protocol→7920842] + GET / deployments); 8c0521d Tier-1 podman container e2e (PROVEN: control-plane container boots, ota-device-emu container runs the real round-trip — evidence docs/qa/20260610T105306Z/); 5d4920e dashboard ArtifactUpload + populated-detail (Vitest 50, Playwright 20); submodule pins bumped — constitution ba0f702, ota-rollout-engine 7a90912, ota-artifact-validator 77c6b48, containers 845ad45. Tracker live at docs/Issues.md (ATM-003/004 Operator-blocked: Tier-2 Cuttlefish-on-Linux-KVM / Tier-3 RK3588 hardware) + docs/Fixed.md.
- **Just-shipped this session (on main, pushed all 4 upstreams):**
 - 50ef5c6 — feat(api): per-device telemetry filters + group/ members pagination (OpenAPI synced, redocly-clean).
 - b0b8ee2 — chore(dashboard): sync API client to new pagination/filter params (dashboard client lockstep).
 - 7dc3334 — feat(emulator): Tier-1 Go OTA device-emulator + resilience (server/internal/deviceemu + cmd/ota-device-emu; surfaced the telemetry deployment_id protocol gap).
 - fa571b8 — test(dashboard): comprehensive UI testing system (Vitest 43 + Playwright 17 + ally).
 - a839220 — test(qa): autonomous e2e (50/50) + security (39/39) + HelixQA bank.
 - **containers submodule** advanced to 845ad45 — feat(emulator): ota-device-emu runtime image + on-demand boot recipe (pushed to its github origin); parent gitlink bumped in this commit.
- **Branch:** main. All work merges to main; commit + push fan out to all upstreams.
- **Upstreams (4):** github (git@github.com:HelixDevelopment/helix_ota.git, also origin fetch), gitlab (git@gitlab.com:helixdevelopment1/helix_ota.git), gitflic (git@gitflic.ru:helixdevelopment/helix_ota.git), gitverse (git@gitverse.ru:helixdevelopment/helix_ota.git). origin push fans out to all four.
- **No in-flight background PIDs / detached pushes known at handoff time** — verify with git status + check qa-results/push_failures/ per §11.4.88 before assuming clean.

- **Container runtime on this host:** podman (the pgx integration + dev stack boot via the containers submodule on podman). **No host QEMU / no AVD / no nested KVM on this Apple-Silicon host** — see binding constraints.

2. Current PHASE + immediate NEXT + terminal goal

- **PHASE:** Emulator-driven device testing — **Tier-1 in progress** (a podman container running a Go ota-device-emulator that speaks the real ota-protocol to the control plane). See docs/design/EMULATED_DEVICE_TESTING.md.
- **Immediate NEXT (priority order):**
 1. Tier-1 emulator: stand up the podman ota-device-emulator exercising register → update-check → telemetry → delta → rollout → recall against a live control plane, capturing real evidence under docs/qa/<run-id>/.
 2. Remaining NEXT-wave items (hardware/ingest-gated): device-side TUF (gomobile-go-tuf/v2 — gated on real RK3588 .so/JNI measurement); device payload-apply integration (DeltaApplyDecision → update_engine, needs a real device); row-4 richer telemetry fields (duration_ms/bytes_transferred — blocked on UNVERIFIED ingest).
- **Terminal goal (loop stop condition A, §11.4.126):** a new fully-validated-and-verified version (tag) created AND published across all owned submodules + main repo to all 4 upstreams.

3. Binding constraints (do not violate)

- **Anti-bluff covenant §11.4** — every PASS carries positive captured evidence; metadata-only / config-only / absence-of-error / grep-without-runtime PASS forbidden. Tests AND HelixQA Challenges bound equally.
- **Absolute no-force-push §11.4.113** — force-push (--force, --force-with-lease, +<ref>) is STRICTLY forbidden on every repo/submodule. Integrate via fetch → merge-onto-latest-main → fast-forward push.
- **podman-only on this host** — use the containers submodule (vasic-digital/containers, §11.4.76) for any containerized workload; never host-direct emulator/adb/qemu (§11.4.109 guard).
- **Tier-2 Android A/B is host-gated** — real update_engine A/B + AVB/dm-verity auto-rollback needs Cuttlefish on a Linux + nested-KVM box; the Apple-Silicon applehv host cannot run it. NOT structurally impossible (§11.4.112) — host/hardware-gated only.
- **Exact naming/versions:** Go module github.com/HelixDevelopment/helix_ota/server; submodules under submodules/; constitution submodule at constitution/. Toolchain: Go 1.26.2, Gradle 9.5 + Kotlin 2.3.20 (AGP 8.5.2 + plain kotlin.android builds; Kotlin MPP does NOT). No LaTeX/drawio/plantuml/graphviz/docker on host; PDF export needs a LaTeX engine (not installed).
- **Commit/push discipline §2.1 + §11.4.88** — commit + push to ALL 4 upstreams; pushes may run detached.

How a fresh session resumes from this file alone

Given only this file's path, a new agent: reads §0 handoff docs, runs `git fetch --all`, confirms HEAD against §1, picks up the §2 PHASE/NEXT, and works under the §3 constraints toward the terminal goal — zero additional context required.